Software Requirements Specification (SRS)

CrowdVote AI: Multi-Level Election Prediction System using Swarm Intelligence

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1. 🧭 Introduction

1.1 Purpose

This document specifies the requirements for **CrowdVote AI**, a web-based platform that enables users to predict election outcomes using a **multi-level voting system** and **swarm intelligence algorithms**.

1.2 Scope

The system will:

- Support election prediction for **Kerala and other regions
- Allow admins to manage constituencies and candidates
- Enable users to predict winners across multiple phases
- Calculate prediction accuracy and rank users
- Use AI models to generate aggregated predictions

1.3 Definitions

- **Constituency**: Electoral region
- **Prediction**: User-selected candidate
- **Predictability Score (PS)**: User reliability metric
- **Swarm Intelligence**: Collective decision-making model
- **Phase**: Time-based prediction stage

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2. 🧾 Overall Description

2.1 Product Perspective

The system is a **standalone web + mobile platform** with AI integration.

2.2 Product Functions

- User registration and authentication
- Multi-phase prediction submission
- Admin management of election data
- AI-based prediction aggregation
- Scoring and leaderboard generation

2.3 User Classes

User Type	Description
Admin	Manages election data
Registered User	Participates in predictions
Public Viewer	Views predictions and results

2.4 Operating Environment

- Frontend: Web / Mobile browser
- Backend: Cloud-based server
- Database: Relational/NoSQL
- AI Engine: Python-based

2.5 Constraints

- Data privacy compliance
- Real-time scalability
- Election data accuracy
- Ethical neutrality

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3. ⚙️ System Features

3.1 Admin Module

Functional Requirements

- Admin shall:
  - Add/edit/delete Elections
  - Add/edit/delete constituencies
  - Add candidates per constituency
  - Configure election phases
  - Upload election results - Evaluate ProdictionAccuracy
  - Control phase activation

3.2 User Management

- Users shall:
  - Register/login securely
  - View profile and prediction history
  - Track accuracy and ranking

3.3 Multi-Level Voting System

Phases Supported

1. Pre-Election Phase
2. Campaign Phase
3. Final Prediction Phase
4. Exit Prediction Phase
5. Post-Result Analysis

Functional Requirements

- Users shall:
  - Submit predictions for each phase
  - Update predictions during campaign phase
  - Lock predictions in final phase
  - Provide confidence level

3.4 Prediction Engine

Functional Requirements

- System shall:
    - Aggregate predictions
    - Apply weighted scoring
    - Generate constituency-level predictions
    - Generate overall election forecast

Algorithm Requirement

Weighted Prediction Model
Final Candidate Score = Σ (User Prediction × Predictability Score × Phase Weight)

3.5 Predictability Score System

Functional Requirements

- System shall compute PS based on:
    - Accuracy
    - Consistency
    - Early prediction correctness
    - Confidence calibration

3.6 Reward System

- System shall:
    - Identify correct predictors
    - Randomly select winners per constituency
    - Assign rewards and badges

3.7 Analytics Dashboard

User Dashboard

- Prediction accuracy
- Rank
- Phase-wise performance

Admin Dashboard

- Prediction trends
- Swing constituencies
- Participation statistics

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4. 📊 External Interface Requirements

4.1 User Interface

- Responsive web interface
- Mobile-friendly design
- Dashboard visualization

4.2 Software Interfaces

- AI libraries (Python ML frameworks)
- Social media APIs (optional):
    - X
    - Facebook

4.3 Hardware Interfaces

- Standard mobile/desktop devices

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5. 🔐 Non-Functional Requirements

5.1 Performance

- Handle thousands of concurrent users
- Real-time prediction aggregation

5.2 Security

- Secure authentication (JWT/OAuth)
- Data encryption
- Protection against manipulation

5.3 Usability

- Simple prediction interface
- Easy navigation
- Minimal learning curve

5.4 Reliability

- 99% uptime
- Backup and recovery system

5.5 Scalability

- Expandable to national/global elections

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6. 🧠 AI & Algorithm Requirements

6.1 Swarm Intelligence

- System shall simulate:
    - Collective decision-making
    - Dynamic convergence

6.2 Machine Learning Models

- Logistic Regression
- Random Forest
- Bayesian Updating
- Time-series analysis

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7. 🔄 System Workflow

1. Admin configures election
2. Users register and predict
3. Predictions stored per phase
4. AI engine processes data
5. System generates forecasts
6. Results uploaded
7. Accuracy calculated

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8. 📈 Evaluation Metrics

- Prediction Accuracy (%)
- User Participation Rate
- Model Confidence Score
- Convergence Efficiency

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9. ⚠️ Assumptions & Dependencies

- Users provide honest predictions
- Election data is accurate
- Internet connectivity available

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10. 🔮 Future Enhancements

- Sentiment analysis integration
- Geo-based prediction clustering
- AI chatbot for insights
- Integration with real-time news

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11. 📌 Conclusion

The proposed system combines:

- **Multi-level voting**
- **Swarm intelligence**
- **AI-based prediction models**

to create a **dynamic, scalable, and intelligent election prediction platform** for **Kerala and beyond.

⚙️ Functional Requirements

CrowdVote AI – Election Prediction System

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1. 👤 User Management Module

FR-1: User Registration

- The system shall allow users to create an account using:
  - Email and password
  - Mobile number (OTP verification optional)

FR-2: User Authentication

- The system shall allow registered users to log in and log out securely
- The system shall support password reset functionality

FR-3: User Profile Management

- The system shall allow users to:
  - View profile details
  - View prediction history
  - View accuracy and Predictability Score

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2. 🧑‍💼 Admin Module

FR-4: Constituency Management

- The admin shall be able to:
  - Add, edit, and delete constituencies

    - Activate/deactivate constituencies

FR-5: Candidate Management

- The admin shall be able to:
    - Add candidates to each constituency
    - Edit candidate details
    - Remove candidates

FR-6: Election Configuration

- The admin shall be able to:
    - Create elections (e.g., Assembly, Lok Sabha)
    - Assign constituencies to elections
    - Define election timelines

FR-7: Phase Management

- The admin shall be able to:
    - Configure prediction phases:
        - Pre-election
        - Campaign
        - Final
        - Exit
    - Set start and end time for each phase
    - Lock/unlock phases

FR-8: Result Management

- The admin shall be able to:
    - Upload official election results
    - Update constituency winners
    - Trigger accuracy calculation

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3. 🗳️ Prediction Module

FR-9: Prediction Submission

- The system shall allow users to:
    - Predict one candidate per constituency
    - Submit predictions for each phase

FR-10: Prediction Update

- The system shall allow users to:
    - Modify predictions during the campaign phase
    - Restrict modifications after final phase lock

FR-11: Confidence Input

- The system shall allow users to assign a confidence level:
    - Low / Medium / High

FR-12: Bulk Prediction

- The system shall allow users to:
    - Predict all constituencies in a single session

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4. 🔄 Multi-Level Voting System

FR-13: Phase-Based Predictions

- The system shall store predictions separately for each phase

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FR-14: Phase Weighting

- The system shall assign predefined weights to each phase

FR-15: Final Prediction Lock

- The system shall:
    - Prevent edits after the final phase deadline

FR-16: Exit Prediction

- The system shall allow predictions after voting but before results

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5. 🧠 Prediction Engine

FR-17: Aggregation of Predictions

- The system shall:
  - Aggregate user predictions per constituency

FR-18: Weighted Prediction Calculation

- The system shall compute candidate scores using:
  - User Predictability Score
  - Phase weight
  - Confidence level

FR-19: System Prediction Generation

- The system shall:
  - Identify the most probable winner per constituency
  - Generate overall election forecast

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6. 📊 Predictability Score Module

FR-20: Score Calculation

- The system shall compute Predictability Score based on:
  - Accuracy
  - Consistency
  - Early prediction correctness
  - Confidence calibration

FR-21: Score Update

- The system shall update scores after each election

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7. 🏆 Reward & Incentive Module

FR-22: Correct Prediction Identification

- The system shall identify users who predicted correctly

FR-23: Random Winner Selection

- The system shall randomly select winners among correct predictors for each constituency

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FR-24: Reward Allocation

- The system shall assign:
    - Points
    - Badges
    - Leaderboard ranks

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8. 📈 Analytics & Dashboard

FR-25: User Dashboard

- The system shall display:
    - Prediction accuracy
    - Rank
    - Phase-wise performance

FR-26: Admin Dashboard

- The system shall display:
    - Participation statistics
    - Prediction trends
    - Swing constituencies

FR-27: Trend Analysis

- The system shall:
  - Track prediction changes across phases
  - Identify convergence patterns

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9. 🔔 Notification Module

FR-28: Notifications

- The system shall notify users about:
  - Phase start/end
  - Prediction deadlines
  - Results announcement

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10. 🔐 Security & Integrity

FR-29: Data Validation

- The system shall validate all user inputs

FR-30: Fraud Prevention

- The system shall:
  - Prevent duplicate accounts
  - Detect suspicious prediction patterns

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11. 🌐 Public Access Module

FR-31: Public Viewing

- The system shall allow non-registered users to:
  - View aggregated predictions
  - View leaderboards

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12. 🔗 Integration Module

FR-32: External Data Integration

- The system may integrate with:
  - Social media platforms like X
  - News APIs for sentiment analysis

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13. 📂 Data Management

FR-33: Data Storage

- The system shall store:
  - User data
  - Predictions
  - Results
  - Scores

FR-34: Data Backup

- The system shall periodically back up all critical data

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14. 📊 Reporting Module

FR-35: Report Generation

- The system shall generate:
  - Prediction reports
  - Accuracy reports
  - Election insights

📌 Summary

This functional requirements specification ensures that the system:

✅ Supports **multi-level prediction workflows**
✅ Implements **AI-driven aggregation**
✅ Enables **user engagement and gamification**
✅ Provides **scalable election analytics**